

1. Scenario Description: Stopped Vehicle or Debris in the Roadway

This scenario looks at how two trucks driving in a platoon deal with a sudden emergency on the highway. The trucks are only about 15 meters apart, so they rely heavily on their built-in ADAS features, mainly lane keeping, distance monitoring, and automatic emergency braking. They also have a wireless V2V link that lets them share safety information instantly. The idea is that if something unexpected shows up in front of the lead truck, both trucks can react almost at the same time instead of one after the other.

2. Functional Requirements Identified

1. The lead truck's radar and camera system should detect any kind of obstacle with minimum dimensions of $0.5\text{ m} \times 0.5\text{ m}$ at a distance of ≥ 150 meters.
2. When the lead truck initiates emergency braking, it should immediately transmit a V2V emergency brake warning to the trailing truck.
3. When the trailing truck gets the warning, it shall automatically apply emergency braking with a deceleration rate that matches or exceeds the lead truck's deceleration.
4. The trailing truck must continuously monitor the distance to the lead truck.

3. Non-Functional Requirements Identified

1. The V2V communication between trucks needs to be extremely fast. V2V communication latency shall remain below 10 ms.
2. The wireless link must deliver safety-critical hazard warnings with at least 99.99% reliability to prevent accidents.
3. The ADAS sensor fusion system should be able to detect and classify stationary obstacles within about 50 milliseconds.

4. Timing Requirements Identified

1. An obstacle needs to be detected and verified by the system in around 50 milliseconds.
2. A delay of under 10 milliseconds is realistic for V2V communication and necessary to maintain a 15-meter gap safely.
3. The trailing vehicle shall initiate its braking system within 100 ms after receiving information via V2V.
4. The system is required to react in less than 160 ms following detection, communication, and brake activation.

5. Safety Requirements Identified

1. In such scenarios, like sudden or emergency braking, the system must prevent collisions between the lead truck and the trailing truck.
2. Emergency braking must stay within safe deceleration limits so that the truck maintains stability and does not lose control.
3. The system should avoid unnecessary or false emergency braking, since sudden braking at highway speeds can create secondary risks for surrounding traffic.

6. Documented Assumptions

1. All trucks are equipped with V2V communication hardware.
2. All trucks have functional sensors.
3. No major mechanical failures.
4. Roads and weather are normal.
5. The obstacle is fully within the lane and not moving.